

VIDYATRACK - QA TEST SCENARIOS

Format: Manual + Integration + Negative + Expected Failures

ROLE: TEACHER

GLOBAL PRE-CONDITIONS (for all TEACHER tests)

- Auth is OTP-based:
 - POST /api/v1/auth/otp/request
 - POST /api/v1/auth/otp/verify → returns TokenOut.access_token
 - GET /api/v1/auth/me (identity)
 - GET /api/v1/teachers/me (teacher profile)
 - GET /api/v1/teachers/me/teaching-assignments (assignments)
 - GET /api/v1/teachers/me/attendance-section (attendance section)
- Authorization header: Authorization: Bearer <access_token>
- Teacher test accounts:
 - T_ACTIVE: active teacher with assigned section + subject
 - T_INACTIVE: inactive teacher
- Known IDs:
 - section_id_assigned = <int>
 - section_id_unassigned = <int>
 - student_id_in_assigned_section = <int>
 - student_id_not_in_section = <int>
 - subject_id_assigned = <int>
 - subject_id_unassigned = <int>

1) Feature: Teacher Authentication & Identity

User Story: US-AUTH-01 Teacher Login via OTP

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A) HAPPY PATH (Manual/UI)

1. Enter phone → Request OTP
2. Enter OTP → Verify
3. Land on teacher home → verify teacher identity + assignments visible

B) INTEGRATION (API)

1) Request OTP

POST /api/v1/auth/otp/request

Payload:

```
{ "phone": "9876543210" }
```

Expected:

```
200 { "status": "<string>" }
```

2) Verify OTP

POST /api/v1/auth/otp/verify

Payload:

```
{ "phone": "9876543210", "otp": "123456" }
```

Expected:

```
200 { "access_token": "<jwt>", "token_type": "bearer" }
```

3) Validate identity

GET /api/v1/auth/me

Expected: 200 (schema currently {} in spec; still validate 200 and non-empty body if implemented)

4) Validate teacher profile

GET /api/v1/teachers/me

Expected 200:

```
{ "teacher_id": <int>, "name": "<string>", "email": "<string|null>", "phone": "<string|null>" }
```

5) Validate assignments

GET /api/v1/teachers/me/teaching-assignments

Expected 200: array of TeachingAssignmentOut

C) NEGATIVE TESTS

NEG-AUTH-01.1 Inactive teacher login blocked

- Attempt verify OTP for T_INACTIVE

Expected:

- 403 Forbidden OR 401 Unauthorized (implementation choice) with message like:

```
{ "detail": "Inactive user" }
```

Must NOT:

- return access_token

NEG-AUTH-01.2 Invalid OTP

Expected:

- 401 Unauthorized or 422 (if validation-driven)

NEG-AUTH-01.3 Missing token on protected endpoint

- Call GET /api/v1/teachers/me with no Authorization header

Expected:

- 401 Unauthorized

Risk checks:

- Token cannot access another teacher's data (RBAC + tenant)

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2) Feature: Daily Attendance Management

User Stories: US-ATT-01, US-ATT-02, US-ATT-03

OpenAPI alignment: AttendanceCreate, AttendanceOut, AttendanceSubmitIn, AttendanceSubmissionOut

Endpoints: POST/GET /api/v1/attendance, POST /api/v1/attendance/submit

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US-ATT-01: Daily Attendance Entry

A) HAPPY PATH (Manual/UI)

1. Login as T_ACTIVE
2. Open Attendance (assigned section auto-loaded from /teachers/me/attendance-section)
3. Mark attendance for each student
4. Submit
5. Refresh → verify attendance remains recorded for date

B) INTEGRATION (API) – Create Attendance rows (per student)

POST /api/v1/attendance

Payload (AttendanceCreate):

```
{ "student_id": 101, "date": "2026-01-22", "status": "PRESENT" }
```

Expected 201 (AttendanceOut):

```
{  
  "id": <int>,  
  "school_id": <int>,  
  "student_id": 101,  
  "date": "2026-01-22",  
  "status": "PRESENT",  
  "marked_by_user_id": <int|null>
```

}

C) INTEGRATION (API) – List Attendance for a date (validation)

GET /api/v1/attendance?date=2026-01-22

Expected 200: array of AttendanceOut

Validate:

- student_id present for submitted students
- date matches
- statuses match

D) INTEGRATION (API) – Submit Attendance for section/date

POST /api/v1/attendance/submit

Payload (AttendanceSubmitIn):

```
{ "section_id": <section_id_assigned>, "date": "2026-01-22" }
```

Expected 201 (AttendanceSubmissionOut):

```
{  
  "id": <int>,  
  "school_id": <int>,  
  "section_id": <section_id_assigned>,  
  "date": "2026-01-22",  
  "submitted_by_user_id": <int|null>  
}
```

E) NEGATIVE TESTS (P0)

NEG-ATT-01.1 Duplicate per-student per-date prevented

- Repeat POST /api/v1/attendance with same student_id + same date

Expected:

- 409 Conflict OR 400 (implementation)

Must NOT:

- create second row

NEG-ATT-01.2 Teacher marks attendance for student outside assigned section

Payload:

```
{ "student_id": <student_id_not_in_section>, "date": "2026-01-22", "status": "PRESENT" }
```

Expected:

- 403/404 or 400 with detail indicating not in teacher scope

NEG-ATT-01.3 Missing required fields

Payload:

```
{ "student_id": 101, "status": "PRESENT" }
```

Expected:

- 422 HTTPValidationError

NEG-ATT-01.4 Invalid date format

Payload:

```
{ "student_id": 101, "date": "22-01-2026", "status": "PRESENT" }
```

Expected:

- 422

NEG-ATT-01.5 Invalid status value

Payload:

```
{ "student_id": 101, "date": "2026-01-22", "status": "X" }
```

Expected:

- 422 or 400 (NOTE: schema does not define enum; if backend allows any string, this is a compliance/data-integrity gap)

US-ATT-02: Same-Day Attendance Editing

NOTE: OpenAPI does NOT show an explicit edit endpoint (PUT/PATCH). Editing may be implemented as:

- re-POSTing attendance and overwriting, OR
- a hidden PATCH endpoint not in spec.

QA requirement: confirm the actual behavior.

A) HAPPY PATH (Manual/UI)

1. Submit attendance for today
2. Edit same day → save

Expected:

- final attendance reflects edits
- edit is auditable (log exists)

B) INTEGRATION (API) – If edit is overwrite via POST

- Re-send POST /api/v1/attendance with same student_id/date but different status

Expected:

- either 200 OK updating existing OR 201 replaced (should not duplicate)

Audit expectation:

- change is logged (where to verify depends on system logging)

C) NEGATIVE

NEG-ATT-02.1 Edit after day ends blocked

- Attempt edit after date boundary

Expected:

- 403/409/400 (implementation)

Must NOT:

- change saved

NEG-ATT-02.2 Teacher without mark_attendance permission

- Attempt POST /api/v1/attendance

Expected:

- 403 Forbidden

US-ATT-03: Attendance Notifications to Parents (triggered automatically)

OpenAPI path for notification trigger is indirect:

- Worker run endpoint exists: POST /api/v1/notifications/outbox/run

A) INTEGRATION (API) – Worker run

POST /api/v1/notifications/outbox/run

Expected 200:

{ "processed": <int> }

B) NEGATIVE / RELIABILITY

NEG-NTF-ATT-03.1 Ensure teacher cannot manually trigger notifications outside worker

- Validate there is no teacher endpoint to send messages

Expected:

- Only outbox worker endpoint exists (management/ops can run; teacher should be blocked by RBAC if required)

NEG-NTF-ATT-03.2 WhatsApp fail → SMS fallback

Expected:

- Outbox processing marks WhatsApp attempt failed and SMS attempted

NOTE: This requires access to outbox logs/status; not exposed in OpenAPI. If not visible, this is a test observability gap (not a UI request; it's a QA evidence gap).

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3) Feature: Marks & Exam Performance Entry

User Stories: US-MRK-01, US-MRK-02, US-MRK-03

OpenAPI alignment: MarksRecordIn, MarksRecordOut, MarksSubmissionIn, MarksSubmissionOut

Endpoints: POST /api/v1/marks/record, POST /api/v1/marks/submit, GET /api/v1/marks

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US-MRK-01: Enter Marks for Exams

A) HAPPY PATH (Manual/UI)

1. Login as T_ACTIVE
2. Select section + subject (assigned) + exam_type
3. Enter marks for all students
4. Submit
5. Verify marks list loads for that section/subject/exam

B) INTEGRATION (API) – Record marks per student

POST /api/v1/marks/record

Payload (MarksRecordIn):

```
{
  "student_id": 101,
  "subject_id": <subject_id_assigned>,
  "exam_type": "UNIT",
  "marks_obtained": 18,
  "max_marks": 20
}
```

Expected 201 (MarksRecordOut):

```
{
  "id": <int>,
  "school_id": <int>,
  "student_id": 101,
  "subject_id": <subject_id_assigned>,
  "exam_type": "UNIT",
  "marks_obtained": 18,
  "max_marks": 20,
  "recorded_by_user_id": <int|null>
}
```

C) INTEGRATION (API) – List marks for section/subject/exam

GET /api/v1/marks?

section_id=<section_id_assigned>&subject_id=<subject_id_assigned>&exam_type=UNIT

Expected 200: array of MarksRecordOut

D) INTEGRATION (API) – Submit marks

POST /api/v1/marks/submit

Payload (MarksSubmissionIn):

```
{ "section_id": <section_id_assigned>, "subject_id": <subject_id_assigned>, "exam_type":
"UNIT" }
```

Expected 201 (MarksSubmissionOut):

```
{
  "id": <int>,
  "school_id": <int>,
  "section_id": <section_id_assigned>,
  "subject_id": <subject_id_assigned>,
  "exam_type": "UNIT",
  "submitted_by_user_id": <int|null>,
  "status": "<string>"
}
```

E) NEGATIVE TESTS (P0)

NEG-MRK-01.1 Teacher attempts record marks for unassigned subject/section

- Use subject_id_unassigned or a student from another section

Expected:

- 403/404

NEG-MRK-01.2 marks_obtained > max_marks

Payload:

{ ..., "marks_obtained": 25, "max_marks": 20 }

Expected:

- 400/422

If accepted, classify as DATA INTEGRITY GAP.

NEG-MRK-01.3 Missing required field

- omit max_marks

Expected: 422

NEG-MRK-01.4 exam_type invalid (schema is string, not enum)

- exam_type="ABC"

Expected:

- should be rejected by backend; if not, DATA INTEGRITY GAP.

US-MRK-02: Mandatory Remarks for Marks

NOTE: OpenAPI does NOT include remarks in MarksSubmissionIn or MarksRecordIn.

This is a spec-vs-BRD gap that QA must flag.

QA Approach (non-architectural):

- Verify UI/API actual includes remarks in some endpoint not captured, or acceptance criteria is not implemented.

Tests:

- Attempt to submit marks without remarks:

Expected (per BRD): blocked.

Observed (per OpenAPI): no remarks field exists → likely cannot be enforced.

Flag as: REQUIREMENT/CONTRACT GAP (P0 if pilot requires it).

US-MRK-03: Consolidated Exam Results to Parents

- Similar evidence gap as notifications:

- No explicit "consolidated results" endpoint in OpenAPI

- Must be outbox-driven worker logic

QA checks:

- Ensure worker sends only after all subjects for exam completed (requires outbox evidence/logs)

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4) Feature: Student Notes & Academic Continuity

User Story: US-NOT-01 Add Student Notes

OpenAPI: StudentNoteCreate, StudentNoteOut

Endpoint: POST /api/v1/students/{student_id}/notes, GET /api/v1/students/{student_id}/notes

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A) HAPPY PATH (Manual/UI)

1. Teacher opens a student profile (from assigned section)
2. Adds a note
3. Note appears in list with timestamp and author attribution

B) INTEGRATION (API)

POST /api/v1/students/101/notes

Payload:

{ "note_text": "Needs improvement in handwriting." }

Expected 201 (StudentNoteOut) includes:

- id, school_id, student_id, author_user_id, note_text, created_at (author_name/role may be null)

C) NEGATIVE TESTS (P0)

NEG-NOT-01.1 Teacher attempts note for unassigned student

- Use student_id outside assigned section

Expected: 403/404

NEG-NOT-01.2 Empty note_text

Payload: { "note_text": "" }

Expected: 422 or 400 (if accepted, DATA QUALITY GAP)

NEG-NOT-01.3 Edit/delete not possible (append-only)

- Attempt PUT/PATCH/DELETE (no endpoints)

Expected: 404 (and UI should not provide such actions)

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5) Feature: Teacher Scope & Access Control

User Story: US-SEC-01 View Own Class and Assignments

OpenAPI: /api/v1/teachers/me, /api/v1/teachers/me/teaching-assignments,
/api/v1/teachers/me/attendance-section

=====

A) HAPPY PATH

- Teacher sees only their assigned section and subjects.

B) INTEGRATION

GET /api/v1/teachers/me/attendance-section

Expected: TeacherAttendanceSectionOut (section_id/name + class_id/name)

GET /api/v1/teachers/me/teaching-assignments

Expected: array of assignments with section + subject

C) NEGATIVE (P0)

- Attempt to access other teacher's resources by changing IDs in other endpoints (students/attendance/marks)

Expected: 403/404

6) Feature: Reliability & Intermittent Internet Handling

User Story: US-SYS-01 Safe Submission Under Connectivity Issues

Manual reliability drills (P0):

REL-SYS-01.1 Double-click submit attendance → must not duplicate

REL-SYS-01.2 Refresh during submit → must not partially save

REL-SYS-01.3 Slow network (simulate throttling) → must not show success unless committed

REL-SYS-01.4 Retry same request after timeout → must be idempotent (no duplicates)

API checks (evidence):

- Re-run same POST /api/v1/attendance/submit (same section/date) → should not create multiple AttendanceSubmissionOut
- Re-run same POST /api/v1/marks/submit (same section/subject/exam) → should not duplicate

KNOWN BRD ↔ OPENAPI GAPS (QA MUST FLAG)

1) Remarks mandatory for marks submission (US-MRK-02)

- OpenAPI has no remarks field in MarksSubmissionIn/MarksRecordIn.
- Action: confirm actual implementation; if absent, log as P0 requirement gap.

2) Same-day attendance edit endpoint not explicit

- OpenAPI provides create + submit; no PATCH/PUT.
- Action: verify actual edit mechanism and enforce same-day rule; otherwise P0 gap.

3) Notification evidence/observability

- OpenAPI exposes only outbox worker run, not message delivery status per recipient.
- Action: confirm logs/status visibility for QA evidence; otherwise P1 evidence gap.

END – ROLE: TEACHER QA TEST SCENARIOS